

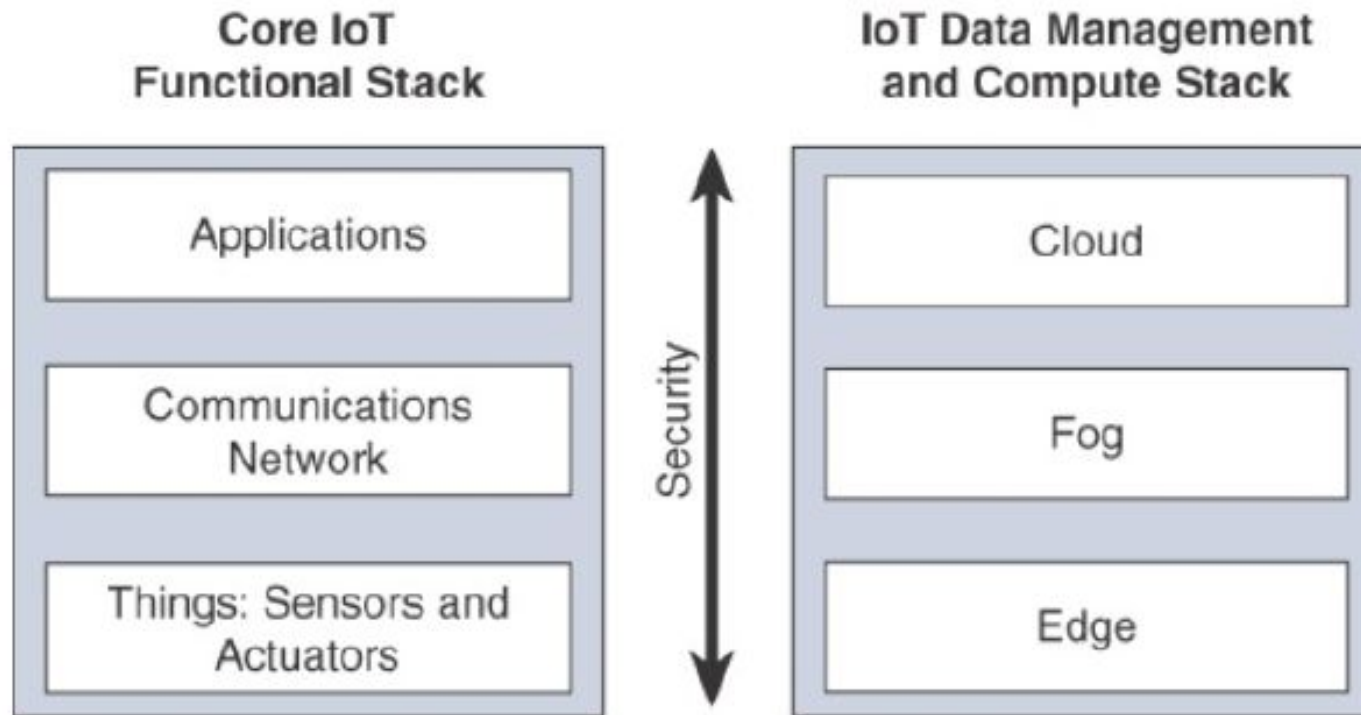
CMP370 –Internet of Things

Unit 2-Architecture and communication Protocols

IoT ARCHITECTURES

- The building blocks of IoT are sensory devices, remote service invocation, communication networks, and context-aware processing of events
- A holistic system architecture for IoT needs to guarantee flawless operation of its components:
- **Reliability** is considered as the most important design factor in IoT and link the physical and virtual realms together.
- **Failure recovery and scalability**
- **Adaptability** to properly handle dynamic interactions within the whole ecosystem.

A Simplified IoT Architecture



- A simplified architecture incorporating two parallel stacks: the Core IoT Functional Stack and the IoT Data Management and Compute Stack.
- The simplified three-layer architecture isn't tied to a specific proposal but evolved organically in the early 2010s as IoT gained traction. It became a common framework in academic literature and introductory IoT courses.

The Core IoT Functional Stack has three layers

- Things layer (also called, Physical Layer/Device Layer/Perception Layer)
- Communication Network layer
- Application and analytics layer

Things (Physical Layer/Device Layer/Perception Layer)

- The process starts with sensors and actuators, the connected devices that monitor (in the case of sensors) or control (in the case of actuators)
- Sensors capture data regarding the status of a process or an environmental condition, such as temperature, humidity, chemical composition, fluid levels in a tank, fluid flow in a pipe, or the speed of an assembly line as well as much more.
- In some cases, a sensor might detect a condition or event that requires an almost immediate response so that an actuator can perform remediation actions in real time, for example, adjusting the flow rate of a fluid, or the movements of an industrial robot. In these situations, very low latency between the sensor and analysis/triggered actuator is required.

Things (Physical Layer/Device Layer/Perception Layer)

- To avoid the delay of a round-trip of data to the server, analysis of data to determine failure and sending of control to the “thing”, this critical processing is performed in close proximity to the process being monitored or controlled.
- This “edge” processing can be performed by a system on module (SOM) device

Communication Network

- When smart objects are not self-contained, they need to communicate with an external system. In many cases, this communication uses a wireless technology.
- A data acquisition system (DAS) collects raw data from the sensors and converts it from analog into digital format.
- The DAS then aggregates and formats the data before sending it through an Internet gateway via wireless WANs (such as Wi-Fi or Cellular) or wired WANs for the next stage of processing.
- At this point, the volume of data is at its maximum. The quantities can be huge, especially, for example, in a factory setting where hundreds of sensors may be gathering data simultaneously.
- For that reason, the data is also filtered and compressed to an optimum size for transmission

Communication Network

- Internet/Network gateways, Data Acquisition System (DAS) are present in this layer.
- DAS performs data aggregation and conversion function (Collecting data and aggregating data then converting analog data of sensors to digital data etc).
- Advanced gateways which mainly opens up connection between Sensor networks and Internet also performs many basic gateway functionalities like malware protection, and filtering also sometimes decision making based on inputted data and data management services, etc.

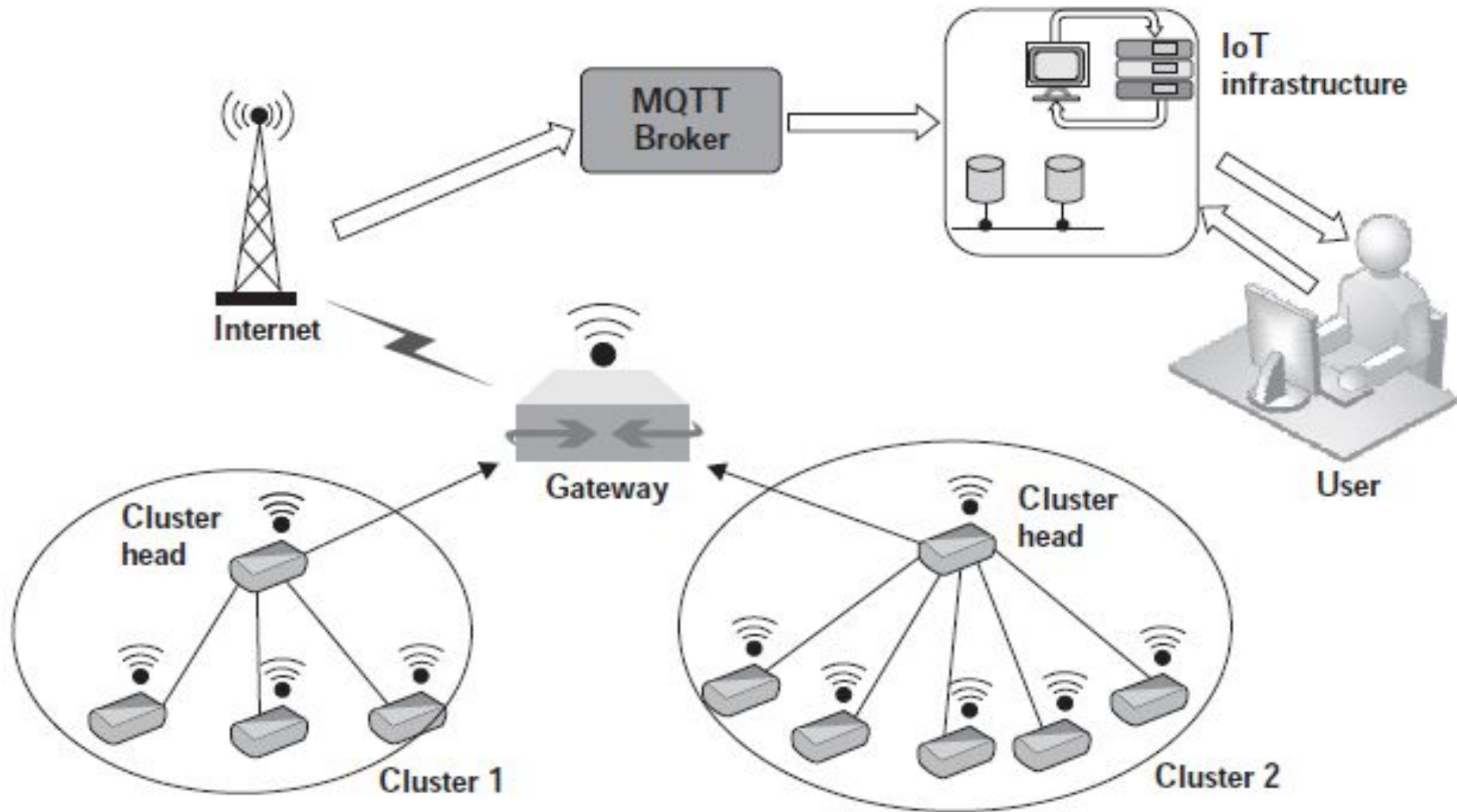
Applications

- At the upper layer, an application needs to process the collected data, not only to control the smart objects when necessary, but to make intelligent decision based on the information collected and, in turn, instruct the “things” or other systems to adapt to the analyzed conditions and change their behaviors or parameters.
- The application layer is what the users see. This could be an application to control a device in a smart-home ecosystem, or a dashboard showing the status of the devices which are part of a system.
- Data centers or cloud is management stage of data where data is managed and is used by end-user applications like agriculture, health care, aerospace, farming, defense, etc.

The IoT Data Management and Compute Stack

- The IoT Data Management and Compute Stack deals with how and where data is filtered, aggregated, stored, and analyzed. In traditional IT models, this occurs in the cloud or the data center.
- However, due to the unique requirements of IoT, data management is distributed as close to the edge as possible, including the edge and fog layers.

IOT System Architecture (Generalized)



Standardization of IIOT architecture

- ITU-T Study Group 20 has taken up the challenge, providing government, industry, and academia with a unique global platform to collaborate in the development of international IIOT standards.
- This Study Group is building on over 10 years of ITU-T experience in IIOT standardization, to enable the coordinated development of IIOT technologies.
- A central part of this study is the standardization of end-to-end architecture for IIOT and mechanisms for the interoperability of IIOT applications and data sets employed by various vertical industries

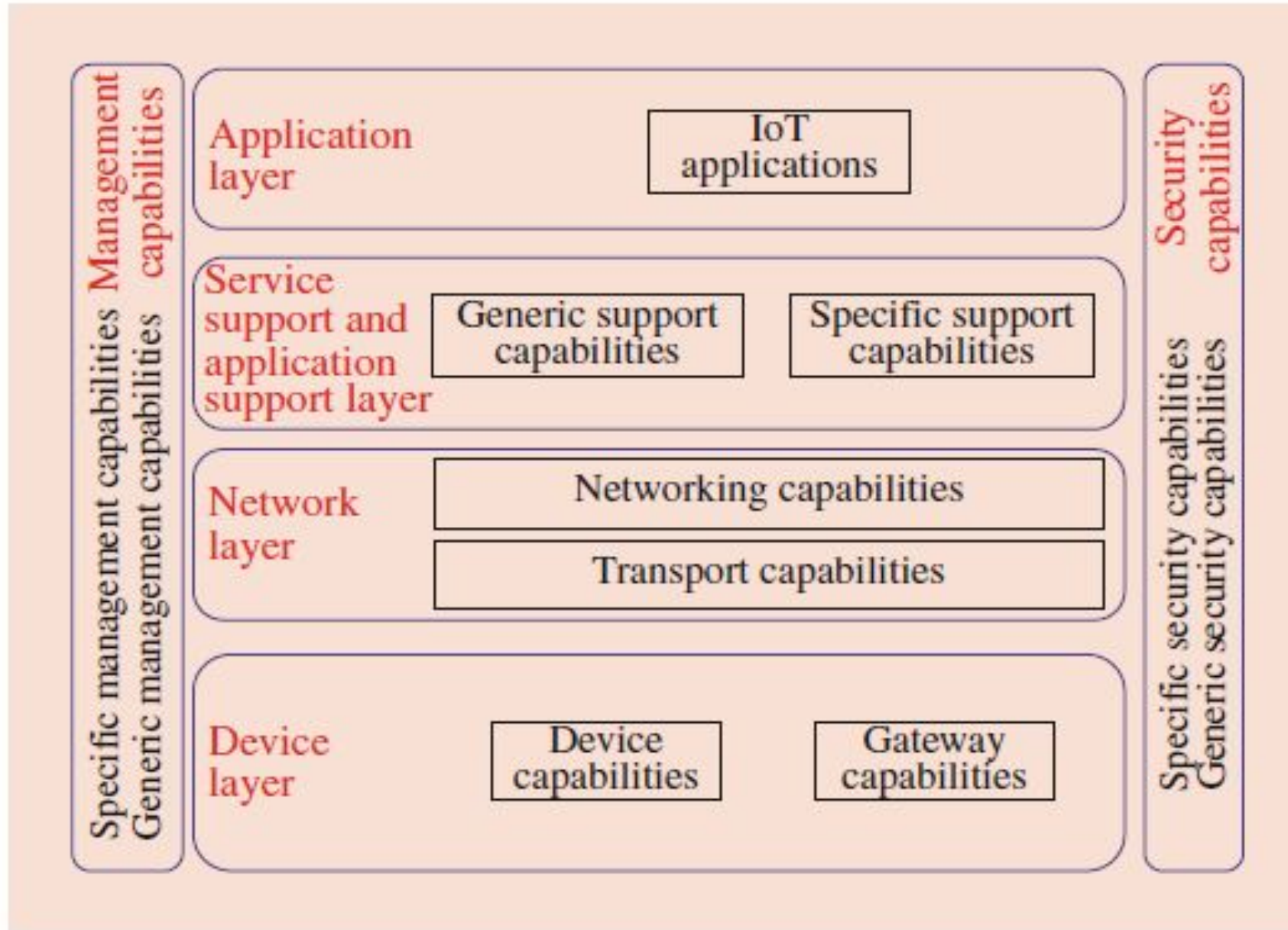
The challenges of IOT Architecture

- IOT system can be extremely diverse, ranging from simple sensor network to complex industrial automation setups.
- This diversity makes it challenging to have a single universally agreed-upon architecture
- The field of IoT is rapidly evolving, with new technologies and applications emerging constantly.
- Lack of Universal Protocols and Data Formats

IOT Reference Model

- Reference architectures and models give a bird's eye view of the whole underlying system
- These are like blueprints or templates for designing systems. They show the essential components and how they should fit together, without necessarily specifying every single detail.
- Their advantage over other architectures relies on providing a better and greater level of **abstraction**, which consequently hides specific constraints and implementation details.
- Objective is simplifying a complex system by focusing on the essential parts and hiding the less important details

IIOT Reference Model (4-Layer)



IOT Reference Model (4-Layer)

- **Formation of ITU-T Study Group 20:** The International Telecommunication Union-Telecommunication Standardization Sector (ITU-T) established Study Group 20 (SG20) in June 2015 to focus on IoT and its applications, including smart cities and communities.
- SG20 was assigned with developing standards for IoT architectures, protocols, and interoperability, addressing the growing complexity of IoT systems.
- A reference model has been defined in ITU-T Y.4000.
- It is composed of four layers, as well as management capabilities and security capabilities which are associated with the four layers: application layer, service support and application support layer, network layer, and device layer.

IOT Reference Model (4-Layer)

The application layer contains various IoT applications from different domains, e.g., smart home, smart cities, transportation , and enterprise services.

- This layer utilizes the services and data provided by the Service Support and Application Enablement Layer to build specific IoT solutions for various domains.

The service support and application support layer consists of two capability groupings:

- a) **Generic Support Capabilities:** The generic support capabilities are common capabilities which can be used by different IoT applications, such as data processing or data storage.
- b) **Specific Support Capabilities:** The specific support capabilities are capabilities which cater for the requirements of diversified applications.

IOT Reference Model (4-Layer)

The network layer consists of the networking capabilities which handles connectivity including ad hoc networking and communication protocols (e.g., 6LoWPAN, NB-IoT)

- This layer also takes care of mobility management or authentication, authorization, and accounting. It also provides transport capabilities, e.g., connectivity for the transport of IoT service and application- specific data information, as well as the transport of IoT-related control and management information.

IOT Reference Model (4-Layer)

The device layer (or Perception layer): This layer is responsible for sensing and collecting data from the environment or physical objects.

- This layer's capabilities can be logically categorized into two kinds of capabilities:
 - a) **Device Capabilities:** Devices are able to gather and upload information directly (i.e., without using gateway capabilities) to the communication network and can directly receive information (e.g., commands) from the communication network.
 - b) **Gateway Capabilities:** Gateways are seen as devices with advanced capabilities, e.g., support devices connected through different kinds of wired or wireless technologies, such as a controller area network (CAN) bus, ZigBee, Bluetooth, mobile network, or Wi-Fi – depending on the network layer capabilities and deployment.

IOT Reference Model (7-Layer)

IoT World Forum Reference Model

Levels

- 7 Collaboration & Processes**
(Involving People & Business Processes)
- 6 Application**
(Reporting, Analytics, Control)
- 5 Data Abstraction**
(Aggregation & Access)
- 4 Data Accumulation**
(Storage)
- 3 Edge Computing**
(Data Element Analysis & Transformation)
- 2 Connectivity**
(Communication & Processing Units)
- 1 Physical Devices & Controllers**
(The "Things" in IoT)



IOT Reference Model (7-Layer)

The Reference Model introduced in 2014 by Cisco, IBM, and Intel at the 2014 IoT World Forum has as many as seven layers.

Layer 1: Physical Devices and Controllers Layer

- This layer is home to the “things” in the Internet of Things, including the various endpoint devices and sensors that send and receive information. Their primary function is generating data and being capable of being queried and/or controlled over a network.

Layer 2: Connectivity Layer

- The function of this IoT layer is the reliable and timely transmission of data.
- More specifically, this includes transmissions between Layer 1 devices and the network and between the network and information processing that occurs at Layer 3 (the edge computing layer).

IOT Reference Model (7-Layer)

Layer 3: Edge Computing Layer

- At this layer, the emphasis is on data reduction and converting network data flows into information that is ready for storage and processing by higher layers.
- Data aggregation, filtering, and processing at the edge

Layer 4: Data accumulation layer

- Captures data and stores it so it is usable by applications when necessary.

Layer 5: Data abstraction layer

- Data normalization and interoperability across devices and platforms, In other words, Reconciles multiple data formats and ensures consistent semantics from various sources.

IOT Reference Model (7-Layer)

Layer 6: Applications layer

- Interprets data using software applications. Applications may monitor, control, and provide reports based on the analysis of the data.

Layer 7: Collaboration and processes layer

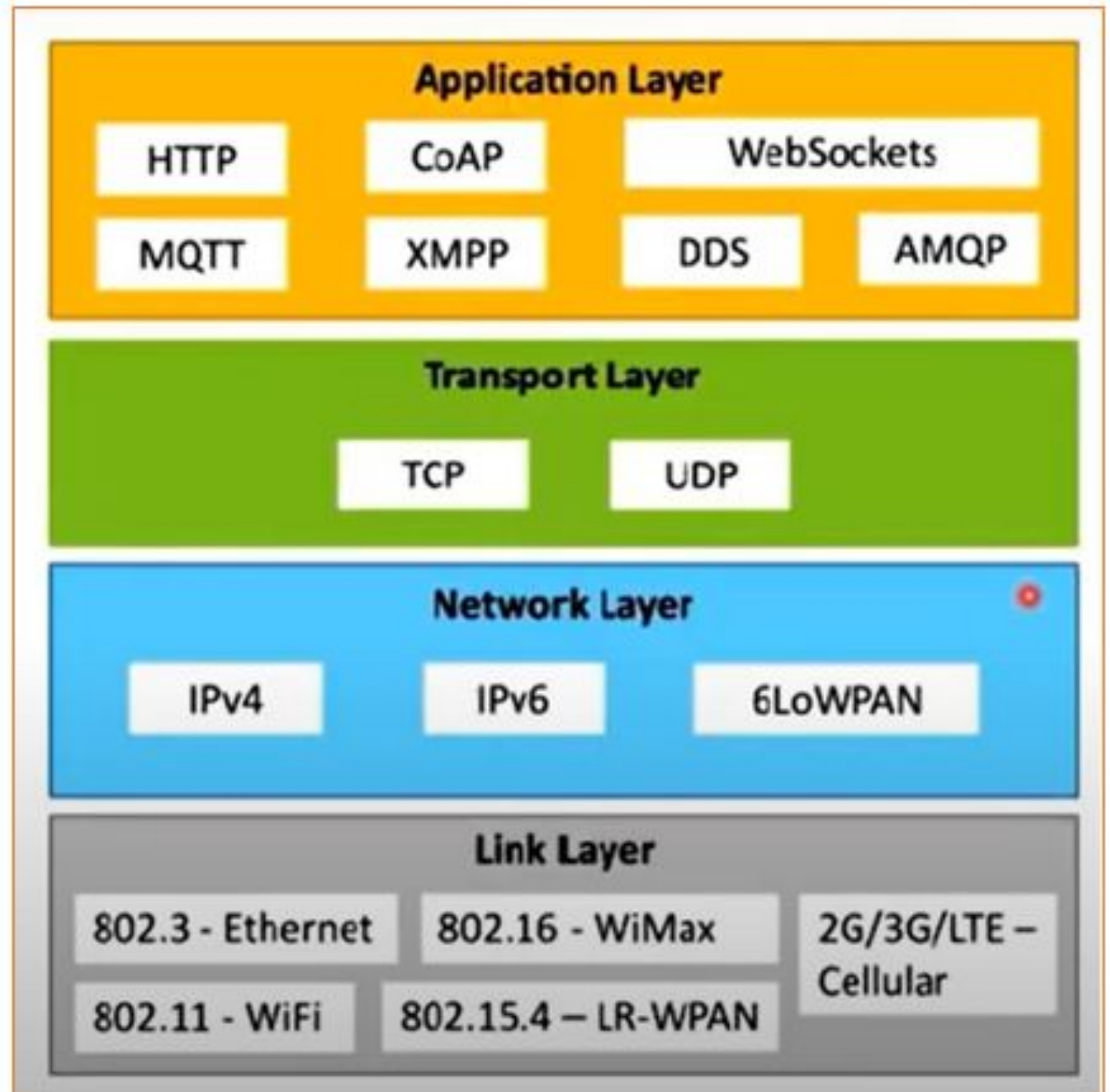
- Consumes and shares the application information.
- Collaborating on and communicating IoT information often requires multiple steps, and it is what makes IoT useful.
- This layer can change business processes, e.g., decision-making based on IoT data and delivers the benefits of IoT.

Aspect	ITU-T Four-Layer Model	IoTWF Seven-Layer Model
Proposer	ITU-T Study Group 20	IoT World Forum (led by Cisco, IBM, Intel)
Initial Proposal	2012 (ITU-T Y.2060, refined by SG20 in 2015)	October 2014 (IoTWF 2014, Chicago)
Focus	Functional capabilities, interoperability	Enterprise IoT, edge computing, IT-OT integration
Development Context	Telecom standardization, smart cities	Industry collaboration, enterprise adoption
Timeline	2012 (Y.2060), 2015 (SG20 formed), 2016-2018 (refinements)	2013 (IoTWF launched), 2014 (model introduced), 2015+ (adoption)
Complexity	Simpler, four layers	More complex, seven layers

Architecture (Optional/Self-Study)

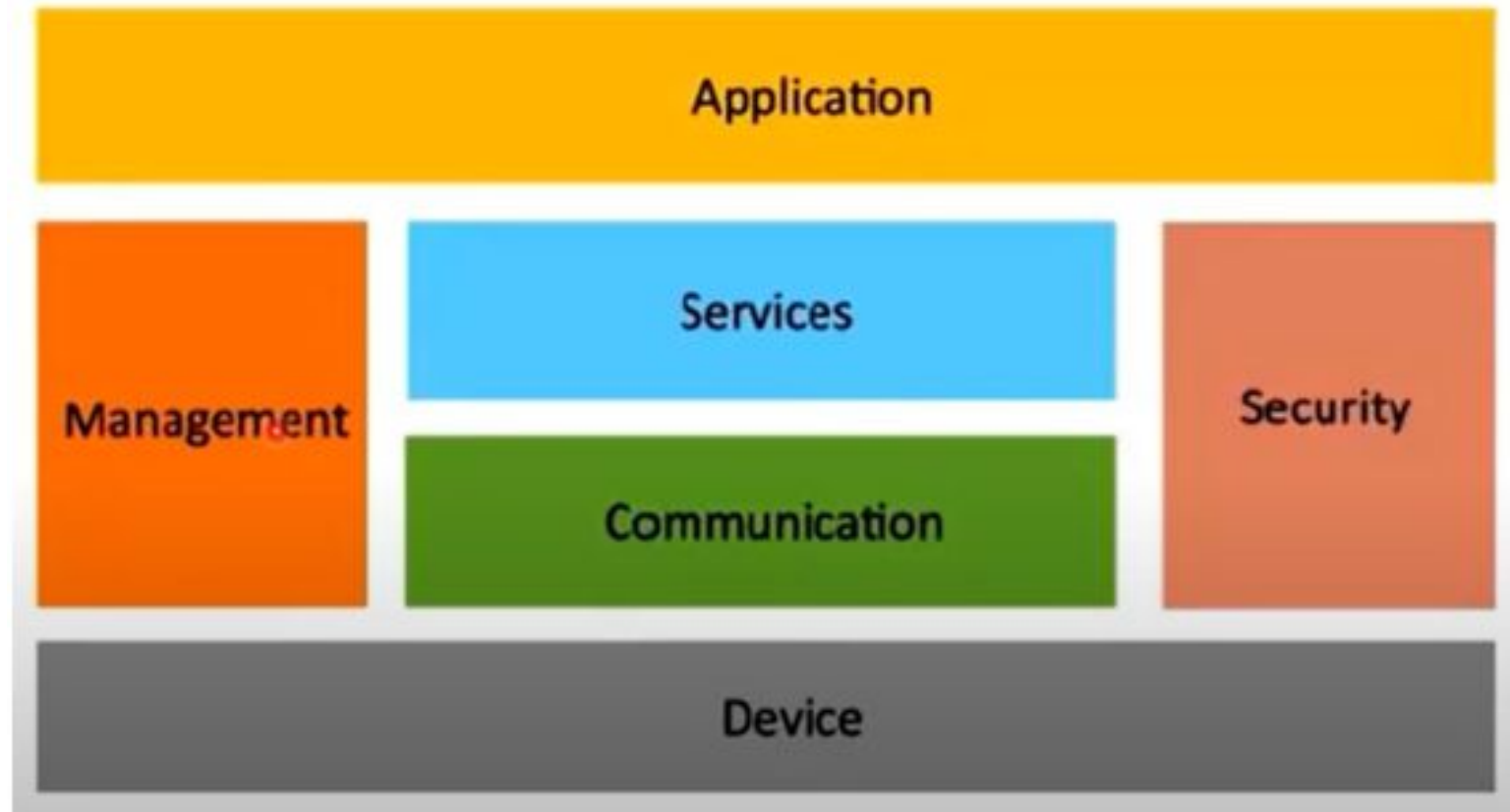
- SOA-BASED ARCHITECTURE
 - service-oriented architecture (SOA)
- API-ORIENTED ARCHITECTURE

IoT Protocols



Logical Design of IoT: Functional Block

- An IoT system comprises of number of functional blocks that provide the system the capabilities for identification, sensing, actuation, communication and management



Logical Design of IoT: Functional Block

- **Device**
 - IoT system devices provides sensing, actuation, monitoring and control function
- **Communication**
 - Handles communication for IOT system
- **Services**
 - Provides services for device monitoring, device control service, data publishing services and service for discovery
- **Management**
 - It provides a various functions to govern the IoT System . E.g. Data organization, data storage

Logical Design of IoT: Functional Block

- **Security**

- This block secures the IoT system by providing function such as authentication, authorization, message and content integrity and data security

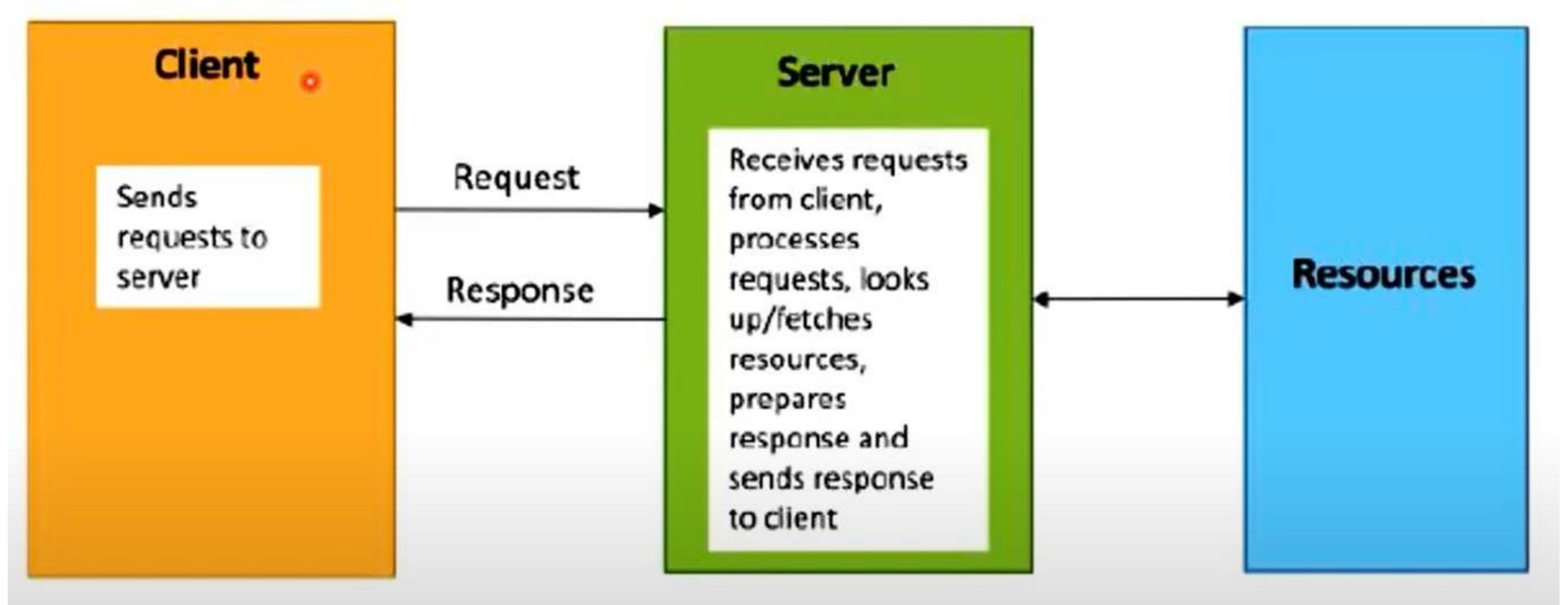
- **Application**

- This is an interface that the users can use to control and monitor various aspects of the IoT system
- It also allows user to view the system status and view or analyze the processed data

Communication Models

- Request-Response Model
- Publish- Subscribe Model
- Push-pull Model
- Exclusive pair model

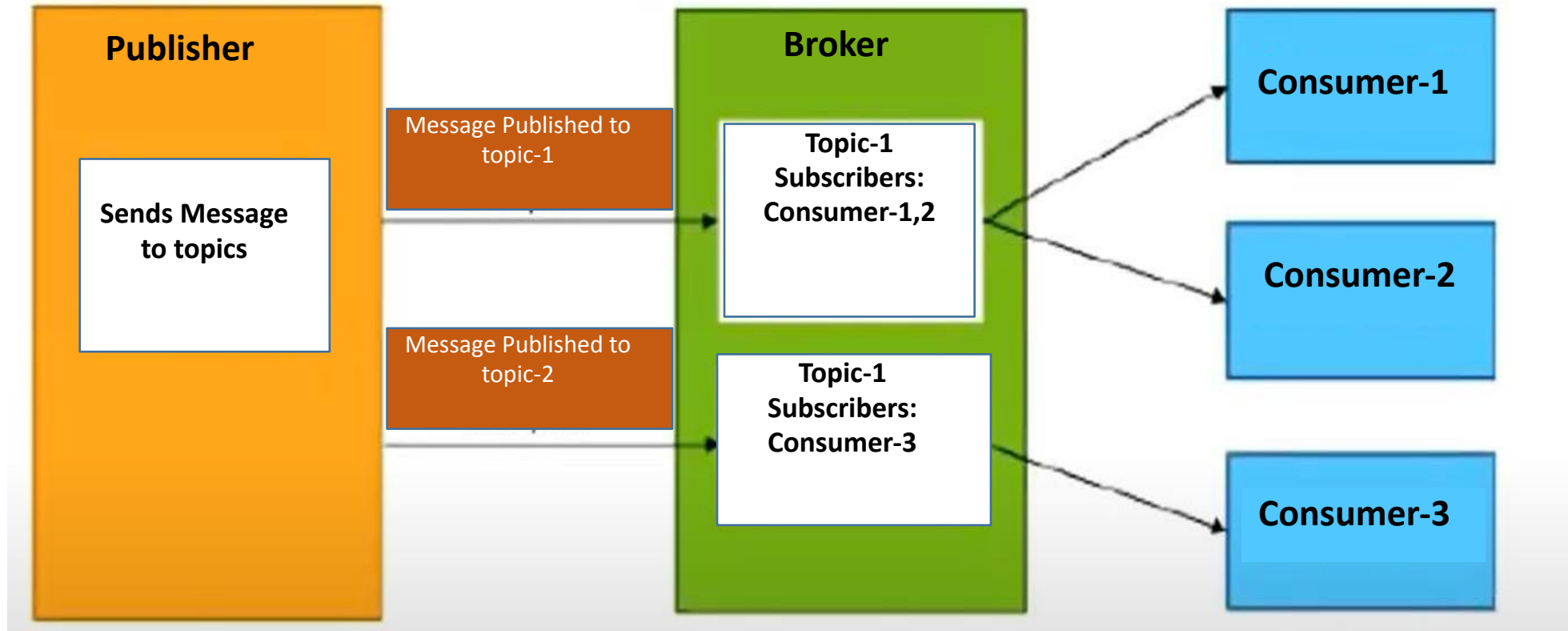
Request-Response Model



Request-Response Model

- The client sends request to the server and the server responds to the request
- The request may be for transfer of data or upload of data
- The server may be remote or local and can handle request of multiple client
- When a server receives a request, it decides how to respond, fetch the data, retrieves resource representation , prepares the response and then sends response to the client
- E.g. HTTP – stateless communication model (Web browser and web server)

Publish-Subscribe Model



Publish- Subscribe Model

This communication model involves: Publisher, Broker and Consumer

- **Publisher** : Source of data , they send the data to the topics which are managed by the broker
- Publisher are not aware of the consumer
- **Consumer**: Subscribe to the topics which are managed by the broker
- **Broker**: when a broker receives a data for a topic from publisher it sends the data to the subscribed consumer . Broker accepts data from publisher and send it to the appropriate consumer

IoT Protocols

Link Layer

- Link layer protocols determine how the data is physically sent over the network's physical layer or medium (e.g., copper wire, coaxial cable, or a radio wave).
- The scope of the link layer is the local network connection to which host is attached. Hosts on the same link exchange data packets over the link layer using link layer protocols.
- Link layer determines how the packets are coded and signaled by the hardware device over the medium to which the host is attached (such as a coaxial cable).

Link layer protocols

- **802.3 - Ethernet** : IEEE 802.3 is a collection of wired Ethernet standards for the link layer.
- **802.11 - WiFi** : IEEE 802.11 is a collection of wireless local area network (WLAN) communication standards
- **802.16 - WiMax** : IEEE 802.16 is a collection of wireless broadband standards, including extensive descriptions for the link layer (also called WiMax).
- **802.15.4- LR-WPAN** : IEEE 802.15.4 is a collection of standards for low-rate wireless personal area networks (LR-WPANS).
- **2G/ 3G/ 4G - Mobile Communication**

Network/Internet Layer

The network layers are responsible for sending of IP datagrams from the source network to the destination network. This layer performs the host addressing and packet routing. The datagrams contain the source and destination addresses which are used to route them from the source to destination across multiple networks.

- **IPv4** : Internet Protocol version 4 (IPv4) is the most deployed Internet protocol that is used to identify the devices on a network using a hierarchical addressing scheme. IPv4 uses a 32-bit address scheme
- **IPv6** : Internet Protocol version 6 (IPv6) is the newest version of Internet protocol and successor to IPv4. IPv6 uses 128-bit address scheme
- **6LoWPAN** : 6LoWPAN (IPv6 over Low power Wireless Personal Area Networks) brings IP protocol to the low-power devices which have limited processing capability.

Transport Layer

The transport layer protocols provide end-to-end message transfer capability independent of the underlying network. The message transfer capability can be set up on connections, either using handshakes (as in TCP) or without handshakes/acknowledgements (as in UDP).

- **TCP** : Transmission Control Protocol (TCP) is the most widely used transport layer protocol, that is used by web browsers (along with HTTP, HTTPS application layer protocols), email programs (SMTP application layer protocol) and file transfer (FTP).
 - TCP ensures reliable transmission of packets in-order.
- **UDP**: UDP is a connectionless protocol.
 - UDP does not provide guaranteed delivery, ordering of messages and duplicate elimination.

Application Layer

- Application layer protocols define how the applications interface with the lower layer protocols to send the data over the network.
- The application data, typically in files, is encoded by the application layer protocol and encapsulated in the transport layer protocol which provides connection or transaction oriented communication over the network.
- Port numbers are used for application addressing (for example port 80 for HTTP, port 22 for SSH, etc.).

Application layer Protocols

- **MQTT (Message Queuing Telemetry Transport)**: Lightweight, publish-subscribe protocol for low-bandwidth, unreliable networks. Used in IoT for real-time data transfer (e.g., sensor data to a cloud server).
 - MQTT is well suited for constrained environments where the devices have limited processing and memory resources and the network bandwidth is low.
- **CoAP (Constrained Application Protocol)**: Constrained Application Protocol (CoAP) is an application layer protocol for machine-to-machine (M2M) applications, meant for constrained environments with constrained devices and constrained networks, uses a request-response model over UDP. Suitable for IoT devices with limited power.
- **HTTP**: Standard web protocol, request-response based, used for IoT applications requiring web integration but heavier than MQTT/CoAP.
- **WebSocket**: Provides full-duplex communication for real-time applications (e.g., live monitoring dashboards). WebSocket is based on TCP and allows streams of messages to be sent back and forth between the client and server while keeping the TCP connection open.